Scheduler

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Brief Introduction:

Scheduler is a system which allow user to efficiently organize events, tasks within events and individual tasks. It allow to add, edit, search and delete task. It keep track of different tasks and events to see dates are not colliding with each other. Task can be prioritized if set within the same date. Task can be tagged (skippable, normal, important) with different categories for faster search, organized structure, and priority value.

Functionalities:

1. User shall be shown dashboard with list of all tasks and events, date wise and priority wise for same date.
2. User shall be notified first, as dashboard appears, if any task is due within 3 days.
3. User shall be able to add, edit, search, mark done and delete task from list.
4. User shall be able to add a daily task with its detail, date to be done by and a tag.
5. User shall be able to open any event.
    a) User shall be able to set opening and closing date of event.
    b) User shall be able to add an event task with its detail, date, and tag.
        i. User shall be able to either queue specific task with other task in the event or add it independently
6. User shall be alerted if any event is collided with any daily task on same date and vice versa while adding any new task or opening new event.
    a) User shall be able to choose either drop/postpone that task/event or leave it within event date.
7. System shall assign a priority value to the task with respect to the tag.
8. User shall be shown task with its priority value and indication.
9. User shall be able to mark done queued task in FIFO behavior keeping track that specified tasks are to be done before task dependent on them.

Data Structures:

List: List for dynamic addition and deletion of the tasks and events.

Hash Maps: Hash Maps for assigning dates and tags to the different tasks for categorizing and faster searching.

Queue or Graph: Queue for provide a FIFO behavior **OR** Graph Topological Sorting for task dependencies, in any event for the list of the task dependent on each other.

Heap: Heaps for tracing priorities of task so that most prioritized tasks are shown top on the dashboard for a specific date.